

STAT4005: Time Series
Tutorial 8 - GARCH Model
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Week 12

8 GARCH Model

8.1 Introduction and Properties of GARCH Model

Define the filtration $\mathcal{F}_t$ as a set that contains all information concerning the time series up to time t .

Heteroskedasticity

Definition 1. A sequence of random variables $\{X_t\}$ is

1. **Heteroskedastic:** if $\text{Var}(X_t)$ varies as t change. Otherwise, $\{X_t\}$ is homoskedastic.
2. **Conditional Heteroskedastic:** if $\text{Var}(X_t|\mathcal{F}_{t-1})$ varies as t changes.

❗ **Remark 1.** In reality, the log-asset price data exhibit the property of volatility clustering, i.e.

$$\text{Var}(X_{t-1}) \text{ is large (or small)} \Rightarrow \text{Var}(X_t|X_{t-1}) \text{ is large (or small)}.$$

This means that a sensible model should induce conditional heteroskedasticity.

Homoskedasticity of Causal ARMA(p, q) Model

Consider $\{X_t\} \sim \text{ARMA}(p, q)$ being causal, then we can write its causal representation by $X_t = Z_t + \sum_{k=1}^{\infty} \psi_k Z_{t-k}$, where $\{Z_t\} \sim \text{WN}(0, \sigma^2)$ and $\sum_{k=1}^{\infty} |\psi_k| < \infty$. Then we have

$$\text{Var}(X_t|\mathcal{F}_{t-1}) = \text{Var}\left(Z_t + \sum_{k=1}^{\infty} \psi_k Z_{t-k}|\mathcal{F}_{t-1}\right) = \text{Var}(Z_t|\mathcal{F}_{t-1}) = \text{Var}(Z_t) = \sigma^2,$$

i.e., the causal ARMA(p, q) model is conditionally and unconditionally homoskedastic, which does NOT match with the properties of the log-asset price data.

It motivates us to consider a conditionally heteroskedastic model, which is known as the GARCH model.

GARCH Model (Generalized Auto-Regressive Conditional Heteroskedastic Model)

Definition 2. We say $\{X_t\} \sim \text{GARCH}(p, q)$ if

$$X_t = \sigma_t \epsilon_t, \quad \{\epsilon_t\} \stackrel{iid}{\sim} \text{N}(0, 1); \quad \sigma_t^2 = \alpha_0 + \sum_{i=1}^p \beta_i \sigma_{t-i}^2 + \sum_{j=1}^q \alpha_j X_{t-j}^2$$

where $\alpha_i, \beta_j \geq 0$ for all i, j . If $p = 0$, we say $\{X_t\} \sim \text{ARCH}(q)$.

Stationarity of GARCH

Theorem 1. If $\{X_t\} \sim \text{GARCH}(p, q)$, then $\{X_t\}$ is weakly stationary if and only if

$$\sum_{j=1}^q \alpha_j + \sum_{i=1}^p \beta_i < 1.$$

❗ **Remark 2.** For GARCH(1, 1) model with $\alpha_1 + \beta_1 = 1$, it is known as the I-GARCH(1, 1) model.

Implication of Stationary GARCH Model

If $\{X_t\} \sim \text{GARCH}(p, q)$ is weakly-stationary, then

- (First Moment of $\{X_t\}$) $E(X_t) = E(\sigma_t \epsilon_t) = E(\sigma_t)E(\epsilon_t) = 0$.
- (Second Moment of $\{X_t\}$) $E(X_t^2) = E(\sigma_t^2 \epsilon_t^2) = E(\sigma_t^2)E(\epsilon_t^2) = E(\sigma_t^2)$.
- (First Moment of $\{\sigma_t^2\}$) $E(\sigma_t^2) = E(\sigma_t^2 \epsilon_t^2) = E(X_t^2) = E(X_{t+k}^2) = E(\sigma_{t+k}^2 \epsilon_{t+k}^2) = E(\sigma_{t+k}^2)$.

Therefore, $E(X_t^2) = E(\sigma_t^2)$ are constant over time.

Then for $\{X_t\} \sim \text{GARCH}(p, q)$ being stationary, we have

Quantity	Unconditional	Conditional on $\mathcal{F}_{t-1}$
Mean	$E(X_t) = 0$	$E(X_t \mathcal{F}_{t-1}) = 0$
Variance	$\text{Var}(X_t) = \frac{\alpha_0}{1 - \left(\sum_{i=1}^q \alpha_i + \sum_{j=1}^p \beta_j \right)}$	$\text{Var}(X_t \mathcal{F}_{t-1}) = \sigma_t^2$
Covariance	$\text{Cov}(X_t, X_{t+h}) = 0$	$\text{Cov}(X_t, X_{t+h} \mathcal{F}_{t-1}) = 0$

Remark 3. We make the following observations

- $\text{Var}(X_t)$ is constant over time, i.e. the model is unconditionally homoskedastic.
- $\text{Var}(X_t | \mathcal{F}_{t-1})$ is not constant over time, i.e. the model is conditionally heteroskedastic.

Basic Properties of GARCH model

 **Exercise 1.** Consider the following GARCH model:

$$X_t = \sigma_t \epsilon_t; \quad \sigma_t^2 = 1.5 + 0.5\sigma_{t-1}^2 + 0.2X_{t-1}^2, \quad \{\epsilon_t\} \stackrel{iid}{\sim} N(0, 1)$$

- Is $\{X_t\}$ stationary?
- Evaluate $E(X_t)$, $\text{Var}(X_t)$, $E(\sigma_t^2)$ and $\text{Cov}(X_t, X_{t+k})$ for $k \neq 0$.
- Compare $\text{Var}(X_t)$ and $\text{Var}(X_t | \mathcal{F}_{t-1})$.

Solution

- As $\alpha_0, \alpha_1, \beta_1 \geq 0$ and $\alpha_1 + \beta_1 = 0.2 + 0.5 = 0.7 < 1$, $\{X_t\}$ is stationary.
- For $\{X_t\}$ being stationary, Both $E(X_t^2) = E(\sigma_t^2)$ is constant over-time and hence

$$E(\sigma_t^2) = 1.5 + 0.5E(\sigma_{t-1}^2) + 0.2E(X_{t-1}^2) = 1.5 + 0.5E(\sigma_t^2) + 0.2E(\sigma_t^2) = 1.5 + 0.5E(\sigma_t^2) + 0.2E(\sigma_t^2).$$
 Hence $E(X_t^2) = E(\sigma_t^2) = 5$. Also, $E(X_t) = E(\sigma_t \epsilon_t) = E(\sigma_t)E(\epsilon_t) = 0$. Then $E(X_t) = 0$, $\text{Var}(X_t) = E(X_t^2) - E(X_t)^2 = 5$. WLOG, assume that $k > 0$ (as $\{X_t\}$ is stationary). Then we have

$$\text{Cov}(X_t, X_{t+k}) = E(X_t X_{t+k}) = E(\sigma_t \sigma_{t+k} \epsilon_t \epsilon_{t+k}) = E(\sigma_t \sigma_{t+k} \epsilon_t) E(\epsilon_{t+k}) = 0.$$
- $\text{Var}(X_t | \mathcal{F}_{t-1}) = \sigma_t^2 = 1.5 + 0.5\sigma_{t-1}^2 + 0.2X_{t-1}^2 \neq 5 = \text{Var}(X_t)$ in general.

ARMA Representation of GARCH Model

Theorem 2. Consider $\{X_t\} \sim \text{GARCH}(p, q)$. If

1. $\{X_t\}$ is stationary;
2. $E(\sigma_t^4)$ exists and is constant over-time

Then $\{X_t^2\} \sim \text{ARMA}(m, p)$ ($m = \max\{p, q\}$) with white noise $\{v_t\}$ and $v_t = \sigma_t^2(\epsilon_t^2 - 1) = X_t^2 - \sigma_t^2$.

Remark 4. The second condition is needed to ensure that $\text{Var}(v_t) < \infty$ is constant. One sufficient condition is to assume $\{\sigma_t^2\}$ being weakly stationary.

Proof of ARMA Representation

Exercise 2. Given that $\{X_t\} \sim \text{GARCH}(p, q)$ is stationary, then express $\{X_t^2\}$ as ARMA process and identify the associated white noise sequence.

Solution

The white-noise sequence is given by $v_t = X_t^2 - \sigma_t^2 = \sigma_t^2(\epsilon_t^2 - 1)$. Then $\{X_t^2\} \sim \text{ARMA}(m, q)$ as

$$\begin{aligned} X_t^2 &= \sigma_t^2 + (X_t^2 - \sigma_t^2) = \alpha_0 + \sum_{j=1}^q \alpha_j X_{t-j}^2 + \sum_{i=1}^p \beta_i \sigma_{t-i}^2 + (X_t^2 - \sigma_t^2) \\ &= \alpha_0 + \sum_{j=1}^m (\alpha_j + \beta_j) X_{t-j}^2 - \sum_{i=1}^p \beta_i (X_{t-i}^2 - \sigma_{t-i}^2) + v_t = \alpha_0 + \sum_{j=1}^m (\alpha_j + \beta_j) X_{t-j}^2 + v_t - \sum_{i=1}^p \beta_i v_{t-i}, \end{aligned}$$

where $\alpha_i := 0$ and $\beta_j := 0$ for $i \geq q, j \geq p$.

Remark 5. Stationary GARCH exhibits some common features with the log-asset series in reality:

- $\{X_t\}$ is heavy-tailed. (Evaluate $E(X_t^4)$)
- $\{X_t^2\}$ is serially correlated; see Theorem 2.
- $\{X_t\}$ is serially uncorrelated; see Ex 1.
- $\{X_t\}$ satisfies volatility cluster. (Definition)

ARMA Representation of GARCH model

 **Exercise 3.** (Continuation of Exercise 1) Consider the following GARCH model:

$$X_t = \sigma_t \epsilon_t; \quad \sigma_t^2 = 1.5 + 0.5\sigma_{t-1}^2 + 0.2X_{t-1}^2, \quad \{\epsilon_t\} \stackrel{iid}{\sim} N(0, 1)$$

Further assume that $E(\sigma_t^4) < \infty$ is constant over time.

- Express $\{X_t^2\}$ as an ARMA model and identify the associated white noise sequence $\{v_t\}$.
- Evaluate $E(\sigma_t^4)$ and $\text{Var}(v_t)$.
- Show that $\{X_t^2\}$ is weakly stationary. Therefore, evaluate $\text{Cov}(X_t^2, X_{t+1}^2)$.

Solution

- (a) The white-noise sequence is given by $v_t = X_t^2 - \sigma_t^2 = \sigma_t^2(\epsilon_t^2 - 1)$. Then

$$\begin{aligned} X_t^2 &= \sigma_t^2 + (X_t^2 - \sigma_t^2) = 1.5 + 0.2X_{t-1}^2 + 0.5\sigma_{t-1}^2 + (X_t^2 - \sigma_t^2) \\ &= 1.5 + (0.5 + 0.2)X_{t-1}^2 - 0.5(X_{t-1}^2 - \sigma_{t-1}^2) + v_t \\ &= 1.5 + 0.7X_{t-1}^2 + v_t - 0.5v_{t-1} \end{aligned}$$

Hence $\{X_t^2\} \sim \text{ARMA}(1, 1)$ with white noise $\{v_t\}$.

- (b) Noticing that $E(\epsilon_t^2) = 1$ and $E(\epsilon_t^4) = 3$. Then

$$\text{Var}(v_t) = E\{(X_t^2 - \sigma_t^2)^2\} - E(X_t - \sigma_t)^2 = E(\epsilon_t^4 \sigma_t^4 - 2\epsilon_t^2 \sigma_t^4 + \sigma_t^4) = 2E(\sigma_t^4).$$

Recall from Exercise 1 that $E(\sigma_t^2) = E(X_t^2) = 5$. Then

$$\begin{aligned} E(\sigma_t^4) &= E\{(1.5 + 0.5\sigma_{t-1}^2 + 0.2X_{t-1}^2)^2\} \\ &= 1.5^2 + 0.5^2 E(\sigma_{t-1}^4) + 0.2^2 E(X_{t-1}^4) + 2(0.5)(0.2)E(\sigma_{t-1}^2 X_{t-1}^2) \\ &\quad + 2(1.5)(0.5)E(\sigma_{t-1}^2) + 2(1.5)(0.2)E(X_{t-1}^2) \\ &= 2.25 + 0.25E(\sigma_t^4) + 0.04E(\epsilon_{t-1}^4 \sigma_{t-1}^4) + 0.2E(\epsilon_{t-1}^2 \sigma_{t-1}^4) + 1.5E(\sigma_{t-1}^2) + 0.6E(\epsilon_{t-1}^2 \sigma_{t-1}^2) \\ &= 2.25 + 0.25E(\sigma_t^4) + 0.04(3)E(\sigma_t^4) + 0.2E(\sigma_t^4) + 1.5(5) + 0.6(5) = 12.75 + 0.57E(\sigma_t^4) \end{aligned}$$

It follows that $E(\sigma_t^4) = 1275/43$ and hence $\text{Var}(v_t) = 2E(\sigma_t^4) = 2550/43$.

- (c) Recall that $X_t^2 = 1.5 + 0.7X_{t-1}^2 + v_t - 0.5v_{t-1}$. As the AR polynomial $\phi(x) = 1 - 0.7x$ gives root $10/7 > 1$, it is causal and therefore weakly stationary. Notice that

- $\text{Cov}(X_t^2, v_t) = \text{Cov}(1.5 + 0.7X_{t-1}^2 + v_t - 0.5v_{t-1}, v_t) = \text{Var}(v_t)$.
- $\text{Cov}(X_t^2, v_{t-1}) = \text{Cov}(1.5 + 0.7X_{t-1}^2 + v_t - 0.5v_{t-1}, v_{t-1}) = 0.2\text{Var}(v_t)$.

Then we have

- $\text{Cov}(X_t^2, \cdot) : \gamma(0) = 0.7\gamma(1) + \text{Var}(v_t) - 0.5 \times 0.2\text{Var}(v_t) = 0.7\gamma(1) + 0.9\text{Var}(v_t)$.
- $\text{Cov}(X_{t-1}^2, \cdot) : \gamma(1) = 0.7\gamma(0) - 0.5\text{Var}(v_t)$.

and hence $\gamma(0) = 55\text{Var}(v_t)/51 = 2750/43$ and

$$\text{Cov}(X_t^2, X_{t+1}^2) = \gamma(1) = 13\text{Var}(v_t)/51 = 650/43.$$

8.2 Estimation and Testing for GARCH Model

In general, we may want to check whether the model is conditional heteroskedastic or not before proceeding.

Lagrange Multiplier Test

Theorem 3. For $\{X_t\} \sim \text{ARCH}(p)$. Consider the regression model $X_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i X_{t-i}^2$ and denote the least-square estimator by $\hat{\alpha}_i$ for $i = 0, \dots, p$. Define

$$\hat{X}_t^2 = \hat{\alpha}_0 + \sum_{i=1}^p \hat{\alpha}_i X_{t-i}^2.$$

Then under $H_0 : \alpha_1 = \dots = \alpha_p = 0$ (i.e., conditional homoskedastic), then $T = nR^2 \xrightarrow{d} \chi_p^2$, where R^2 is the coefficient of determination of the regression model and H_0 is rejected when $T > \chi_{0.95,p}^2$.

We then consider the maximum likelihood estimation of the GARCH model.

Likelihood Estimation for GARCH Model

Theorem 4. For $\{X_t\} \sim \text{GARCH}(p, q)$, define the (conditional) log-likelihood as $\ell(\boldsymbol{\theta} | X_1, \dots, X_n)$, where $\boldsymbol{\theta} = (\alpha_0, \alpha_1, \dots, \alpha_p, \beta_1, \dots, \beta_q)$. Then

1. Assume $\sigma_t^2 = X_t^2 = 0$ for all $t \leq 0$.
2. Evaluate $\sigma_t^2 = \alpha_0 + \sum_{i=1}^p \beta_i \sigma_{t-i}^2 + \sum_{j=1}^q \alpha_j X_{t-j}^2$ for $t = 1, \dots, n$.
3. Write $\ell(\boldsymbol{\theta} | X_{1:n}) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^n \log(\sigma_t^2) - \frac{1}{2} \sum_{t=1}^n \frac{x_t^2}{\sigma_t^2}$

The MLE is thus given by $\hat{\boldsymbol{\theta}} = \arg \max \ell(\boldsymbol{\theta} | X_{1:n})$.

Also, we would like to do model selection. However, there is no general pattern for the ACF and ACVF plots associated with $\text{GARCH}(p, q)$. Therefore, we should use AIC and BIC for model selection.

Information Criterion for GARCH Model

Definition 3. Let $\{X_t\} \sim \text{GARCH}(p, q)$ with $X_1, \dots, X_n$ be observed. Define L as the likelihood for $(X_1, \dots, X_n)$ evaluated under the MLE. Then

1. AIC: $-2 \log L + 2(p + q + 1)$.
2. BIC: $-2 \log L + (p + q + 1) \log n$.

After choosing the desired $\text{GARCH}(p, q)$ model by AIC or BIC, the residual analysis should be carried out to check the goodness of fit of the model. Again, we consider the portmanteau test (Ljung-Box Test)

Ljung-Box Test for GARCH Model

Theorem 5. Let $\{\hat{\epsilon}_t\}$ be the estimated residual of the fitted $\text{GARCH}(p, q)$ model for $\{X_t\}$. Let r_j be the ACF sample of $\{\hat{\epsilon}_t\}$. Then under $H_0 : \text{Cov}(\epsilon_t, \epsilon_{t+k}) = 0$ whenever $|k| \leq h$, the Portmanteau Test (Ljung-Box Test).

$$Q(h) = n(n+2) \sum_{j=1}^h \frac{r_j^2}{n-j} \xrightarrow{d} \chi_{h-p-q-1}^2.$$

Then H_0 is rejected when $Q(h) \geq \chi_{0.95, h-p-q-1}^2$.

Estimation and Testing for GARCH Model

 **Exercise 4.** Given $X_1 = 0$, $X_2 = 4$, $X_3 = 2$ and $X_4 = 3$. Consider fitting $\{X_t\}$ by GARCH(1,1) model, i.e.,

$$X_t = \sigma_t \epsilon_t, \quad \{\epsilon_t\} \stackrel{iid}{\sim} N(0, 1); \quad \sigma_t^2 = \alpha_0 + \beta_1 \sigma_{t-1}^2 + \alpha_1 X_{t-1}^2.$$

(a) Consider the regression model $X_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2$ and the following R-code:

```
x = c(0,4,2,3) ; n = length(x)
fit_LS = lm(x[2:4]^2 ~ x[1:3]^2)
(R2 = summary(fit_LS)$r.squared)
[1] 0.9908257
(T = n*R2)
[1] 3.963303
```

Given $\chi_{0.95,1}^2 = 3.84$ and $\chi_{0.05,2}^2 = 5.99$, test whether there is significant GARCH effect in $\{X_t\}$.

(b) Write down the log-likelihood of (X_1, X_2, X_3, X_4) in terms of $(\alpha_0, \alpha_1, \beta_1)$.

(c) Consider the following R-code:

```
library("tseries")
fit = garch(x, order = c(1,1)) # Fit the data by GARCH(1,1) model
(round(fit$coef, 2))
      a0      a1      b1
2.77 0.00 0.74
```

Suppose $\hat{\sigma}_4^2 = 2$. Find the prediction interval for X_5 .

Solution

(a) $T = nR^2 = 3.96 > 3.84 = \chi_{0.05,1}^2$. Hence we conclude that there is a significant GARCH effect.

(b) Let $\sigma_t^2 = X_t^2 = 0$ for $t \leq 0$. Then

$$\begin{aligned}\sigma_1^2 &= \alpha_0 + \beta_1 \sigma_0^2 + \alpha_1 X_0^2 = \alpha_0 \\ \sigma_2^2 &= \alpha_0 + \beta_1 \sigma_1^2 + \alpha_1 X_1^2 = \alpha_0 + \beta_1 \alpha_0 \\ \sigma_3^2 &= \alpha_0 + \beta_1 \sigma_2^2 + \alpha_1 X_2^2 = \alpha_0 + \beta_1(\alpha_0 + \beta_1 \alpha_0) + 16\alpha_1 \\ \sigma_4^2 &= \alpha_0 + \beta_1 \sigma_3^2 + \alpha_1 X_3^2 = \alpha_0 + \beta_1[\alpha_0 + \beta_1(\alpha_0 + \beta_1 \alpha_0) + 16\alpha_1] + 9\alpha_1\end{aligned}$$

Hence

$$\ell(\alpha_0, \alpha_1, \beta_1 | X_{1:3}) = -\frac{3}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^4 \log(\sigma_t^2) - \frac{1}{2} \sum_{t=1}^4 \frac{X_t^2}{\sigma_t^2}$$

can be written in terms of $(\alpha_0, \alpha_1, \beta_1)$.

(c) As $\hat{\sigma}_5^2 = \hat{\alpha}_0 + \hat{\beta}_1 \hat{\sigma}_4^2 + \hat{\alpha}_1 X_4^2 = 2.77 + 0.74 \times 2 = 4.25$, then $\text{Var}(X_5 | \mathcal{F}_4) \approx 4.25$ and $E(X_5 | \mathcal{F}_4) = 0$, the 95% prediction interval for X_5 is given by

$$(-1.96 \times 4.25^{1/2}, 1.96 \times 4.25^{1/2}) = (-8.33, 8.33).$$

8.3 Additional Exercise

ARCH(∞) Representation

 **Exercise 5.** Consider GARCH(1, 1) model

$$X_t = \sigma_t \epsilon_t, \quad \{\epsilon_t\} \stackrel{iid}{\sim} N(0, 1); \quad \sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

with $\alpha_1 + \beta_1 < 1$. Express the GARCH(1, 1) model as an ARCH(∞) model.

Solution

As $\beta_1 < \alpha_1 + \beta_1 < 1$, we have $(1 - \beta_1 B)\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2$ and hence

$$\begin{aligned} \sigma_t^2 &= (1 - \beta_1 B)^{-1} \alpha_0 + \alpha_1 X_{t-1}^2 = \frac{\alpha_0}{1 - \beta_1} + \alpha_1 \sum_{k=0}^{\infty} \beta_1^k B^k X_{t-1}^2 \\ &= \frac{\alpha_0}{1 - \beta_1} + \alpha_1 \sum_{k=1}^{\infty} \beta_1^{k-1} X_{t-k}^2. \end{aligned}$$

Moments of ARCH(1)

 **Exercise 6.** Assume $\{X_t\} \sim \text{ARCH}(1)$ satisfies

$$X_t = \sigma_t \epsilon_t, \quad \{\epsilon_t\} \stackrel{iid}{\sim} N(0, 1); \quad \sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2$$

with $\alpha_1 < 1$. Assume that $\{\sigma_t^2\}$ is weakly stationary. Evaluate $E(X_t^k)$ for $k = 1, 2, 3, 4$.

Solution

- $E(X_t) = E(\sigma_t \epsilon_t) = E(\sigma_t)E(\epsilon_t) = 0$.
- $E(X_t^2) = E(\sigma_t^2 \epsilon_t^2) = E(\sigma_t^2)E(\epsilon_t^2) = \alpha_0 + \alpha_1 E(X_{t-1}^2) = \alpha_0 + \alpha_1 E(X_t^2) \Rightarrow E(X_t^2) = \alpha_0 / (1 - \alpha_1)$.
- $E(X_t^3) = E(\sigma_t^3 \epsilon_t^3) = E(\sigma_t^3)E(\epsilon_t^3) = 0$.
- Recall that $E(\epsilon_t^4) = 3$. Then

$$\begin{aligned} E(X_t^4) &= E(\sigma_t^4 \epsilon_t^4) = E(\sigma_t^4)E(\epsilon_t^4) = 3E\{(\alpha_0 + \alpha_1 X_{t-1}^2)^2\} \\ &= 3\left(\alpha_0^2 + 2\alpha_0\alpha_1 E(X_{t-1}^2) + \alpha_1^2 E(X_{t-1}^4)\right) \\ &= 3\left(\alpha_0^2 + \frac{2\alpha_0^2\alpha_1}{1 - \alpha_1} + \alpha_1^2 E(X_{t-1}^4)\right). \end{aligned}$$

Hence

$$(1 - 3\alpha_1^2)E(X_t^4) = 3\frac{\alpha_0^2 - \alpha_0^2\alpha_1 + 2\alpha_0^2\alpha_1}{1 - \alpha_1}.$$

It follows that

$$E(X_t^4) = \frac{3\alpha_0^2(1 + \alpha_1)}{(1 - \alpha_1)(1 - 3\alpha_1^2)}.$$

8.4 ✂ Interpretation of GARCH Coefficients

We try to understand more about the GARCH model. For simplicity, consider GARCH(1,1) model, i.e.

$$X_t = \sigma_t \epsilon_t, \quad \{\epsilon_t\} \stackrel{iid}{\sim} N(0, 1); \quad \sigma_t^2 = \omega + \alpha X_{t-1}^2 + \beta \sigma_{t-1}^2$$

with $0 < \alpha + \beta < 1$. Then we have

$$V_L := \text{Var}(X_t) = \frac{\omega}{1 - \alpha - \beta}$$

being constant for all t and hence

$$\sigma_t^2 = (1 - \alpha - \beta)V_L + \alpha X_{t-1}^2 + \beta \sigma_{t-1}^2 \quad \Rightarrow \quad \sigma_t^2 - V_L = \alpha(X_{t-1}^2 - V_L) + \beta(\sigma_{t-1}^2 - V_L).$$

Define $\mathcal{F}_t$ as the set that contains all the information about $(\sigma_t^2, \sigma_{t-1}^2, \sigma_{t-2}^2, \dots)$. (Formally speaking, $\mathcal{F}_t$ is the sigma-algebra generated by $\{\sigma_k^2\}_{k \leq t}$) Then as $E(X_t^2 | \mathcal{F}_t) = E(\sigma_t^2 \epsilon_t^2 | \mathcal{F}_t) = \sigma_t^2 E(\epsilon_t^2 | \mathcal{F}_t) = \sigma_t^2$,

$$\begin{aligned} E(\sigma_t^2 - V_L | \mathcal{F}_{t-1}) &= \alpha \left(E(X_{t-1}^2 | \mathcal{F}_{t-1}) - V_L \right) + \beta(\sigma_{t-1}^2 - V_L) \\ &= (\alpha + \beta)(\sigma_{t-1}^2 - V_L) \\ &= (\alpha + \beta)E(\sigma_{t-1}^2 - V_L | \mathcal{F}_{t-2}) \\ &= \dots = (\alpha + \beta)^k E(\sigma_{t-k}^2 - V_L | \mathcal{F}_{t-k-1}) \end{aligned}$$

It follows that

$$E(\sigma_t^2 | \mathcal{F}_{t-1}) = V_L + (\alpha + \beta)^k E(\sigma_{t-k}^2 - V_L | \mathcal{F}_{t-k-1})$$

and hence

$$\lim_{t \rightarrow \infty} E(\sigma_t^2 | \mathcal{F}_{t-1}) = V_L.$$

Recall that $E(\sigma_t^2) = E(X_t^2) = V_L$ while $E(\sigma_t^2 | \mathcal{F}_{t-1}) \neq V_L$ in general. We then conclude that even though the GARCH model is conditionally heteroskedastic, the conditional variance would still converge to the unconditional variance. Therefore, the quantity V_L is interpreted as **Long-Run Variance** in the GARCH model. A similar result also holds for the general stationary GARCH(p, q) model.

8.5 ✂ Extension of GARCH Model

The following are some well-known extensions of the GARCH model.

Extension of GARCH

Definition 4. Assume $X_t = \sigma_t \epsilon_t$, where $\epsilon_t \stackrel{iid}{\sim} N(0, 1)$.

1. GARCH(1,1): $\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2$.
2. GJR-GARCH(1,1): $\sigma_t^{2\delta} = \alpha_0 + \alpha_1 (|X_{t-1}| - \gamma X_{t-1}^2)^\delta + \beta_1 \sigma_{t-1}^{2\delta}$.
3. EGARCH(1,1): $\log(\sigma_t^2) = \alpha_0 + \alpha_1 |X_{t-1}/\sigma_{t-1}| + \gamma (X_{t-1}/\sigma_{t-1}) + \beta_1 \log(\sigma_{t-1}^2)$.
4. TGARCH(1,1): $\sigma_t^2 = \alpha_0 + \alpha_1 \mathcal{X}_{t-1}^2 + \lambda X_{t-1}^2 \mathbb{1}(X_{t-1} < 0) + \beta_1 \sigma_{t-1}^2$.

Remark 6. Those models have their own aspects and concern in modeling the stochastic volatility process. You can search for their properties if you are interested.